WHITE PAPER · HWACHEON MEGA 100 × OIL & GAS

Hwacheon Mega 100 for Oil & Gas

Heavy-duty CNC lathe with industry-leading 10" thru-hole for tubing, collars, subs, and long-shaft work. This paper covers what the machine does, the oil & gas parts we produce with it, the alloys we run, the tolerances and finishes oil & gas buyers expect, and the documentation package.

16" OD × 100" L × 10" thru-hole
ENVELOPE

±0.0005" routine on critical features
TOLERANCE

1994
SINCE

Same-week / rig-down direct
RESPONSE

Texaco Thunder · 16" OD × 100" L × 10" thru-hole · heavy-duty CNC lathe with 10" thru-hole

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Executive summary

Machine: Hwacheon Mega 100 (shop nickname: *Texaco Thunder*). heavy-duty CNC lathe with 10" thru-hole at B&R Productions, running from our New Waverly, Texas shop.

Application: Heavy-duty CNC lathe with industry-leading 10" thru-hole for tubing, collars, subs, and long-shaft work for Oil & Gas customers — Frac fleet operators, wellhead crews, wireline and coil-tubing service companies, downhole tool manufacturers, completion tool OEMs, subsea equipment vendors, produced-water handling equipment suppliers.

Envelope: 16" OD × 100" L × 10" thru-hole. **Tolerance:** ±0.0005" routine on critical features. **Traceability:** Material by heat and lot on every job.

The machine, in context

The 10-inch through-hole is where Texaco Thunder earns its keep. Long tubing stock, downhole tool bodies with internal passages, coiled-tubing tool bodies, large valve stems with bore access from both ends — parts that other shops outsource because their machines can't accommodate the stock. This machine turns outsourced-work into in-house work.

Why Texaco Thunder for Oil & Gas work

This is the strongest machine-to-industry match in the shop. Oilfield parts are disproportionately tubular and long — downhole tool bodies, coiled-tubing tool bodies, subs, collars, long valve stems — and the constraint that decides who can make them is spindle bore, not swing. A 10" through-hole puts B&R in a small group of shops that can feed the stock through and work it in one piece.

What this looks like on a real part

Take a downhole tool body with an internal profile that has to be accessed from both ends. On a conventional lathe with a 3" bore you cut the stock to length, chuck it short, machine one end, flip it, re-indicate, and machine the other — and every concentricity requirement between the two ends now depends on how well you re-located it. Fed through a 10" bore, the part is worked from both ends without ever losing the original datum. The tolerance stack that would have been the hard part of the job simply doesn't exist.

> **When this is the wrong machine**
>
> Two-range spindle speeds of 400 and 750 RPM mean this is a slow, heavy machine. Small oilfield parts in quantity — seat rings, inserts, bushings — cost more here than they should and belong on Bearkat Blur. Above 16" OD the job moves to Big Thicket Express.

Full specifications

Model	Hwacheon Mega 100 ("Texaco Thunder")
Type	Heavy-duty flatbed horizontal CNC lathe — Mega-100 platform
Max turning capacity	16" OD × 100" L (B&R working spec)
Platform max cutting Ø	27.5" (700 mm) on the Mega-100
Chuck swing over bed	42.5" (1,080 mm)
Distance between centers	10 ft (3,000 mm) B&R config; platform available to 20 ft (6,000 mm)
Spindle bore	10" (255 mm) B&R config; platform range 6" – 12.6"
Spindle speed	400 / 750 RPM (two-range)
Spindle motor	25 HP
Turret	4 or 8-station heavy-duty
Machine weight	~35,000 lb (16,000 kg)
Purpose	Long shafts, main-shaft work, large pipe / tubing components
Tolerance capability	±0.0005" routine on critical features
Materials	Full oilfield alloy range — Inconel, Duplex 2205/2507, 17-4 PH, 4140, Monel

Values marked "platform" reflect the machine manufacturer's published spec range for this platform family; B&R's working spec is what we quote on this specific unit.

Who this serves in Oil & Gas

Typical buyer: Frac fleet operators, wellhead crews, wireline and coil-tubing service companies, downhole tool manufacturers, completion tool OEMs, subsea equipment vendors, produced-water handling equipment suppliers.

What's hard about oil & gas work: Chloride stress-corrosion cracking, sour service (H2S), high-cycle pressure loading, exotic superalloys, and rig-down clocks that turn every hour into money. Documentation trails required for API-adjacent work.

Typical Oil & Gas parts we produce on Texaco Thunder

- Downhole tool bodies with 10" thru-hole bar-stock feed
- Coiled-tubing tool bodies
- Long valve stems with bore access from both ends
- Large tubing spools and hollow-shaft components
- Frac pump plungers and long turned oilfield parts

Above list is representative; if your part isn't shown, call and ask — most oilfield / aerospace / defense / industrial turned or milled work fits somewhere in the shop.

Alloys we run for Oil & Gas

Standard range: Inconel 718 (aged and annealed) · Inconel 625 · Super Duplex 2507 (UNS S32750) · Duplex 2205 · 17-4 PH (all conditions H900 to H1150) · Monel K-500 · Nitronic 50/60 · F22 · F91 · 4130/4140.

Alloy behavior is different in oilfield service. Inconel 718 in the aged condition (H900, roughly 40-45 HRC) is a different machining problem than annealed 718 — feeds, speeds, and tool geometry all shift. Super Duplex 2507 machined without cutting-temperature control develops brittle intermetallics that destroy the chloride-resistance the alloy was chosen for; a shop that shrugs when you ask about phase balance is a shop that will quietly ruin your part. 17-4 PH in Condition A machines like a well-behaved stainless; in H900 it's a different animal. Every one of these alloys we run weekly, not experimentally.

> **What "we run these weekly" means:** we stock or reliably source the common alloys in the industry's typical spec range. Sharp tooling, proven feeds and speeds, and process control specific to each alloy family. Your part is not the one we're experimenting on.

How we machine it — our 5-step process

This is the process B&R runs on every job, oil & gas or otherwise. The specifics of feeds, speeds, and tooling shift by alloy and machine; the discipline is universal.

1. **Material verification.** Every job starts with material traceability. Heat and lot numbers documented against the mill test report. For customer-supplied material, we verify against the CofC before touching the stock.

2. **First-article layout.** Stock is measured and laid out before any cutting starts. Concentricity, roundness, and squareness of the starting material established; if the stock is out of tolerance we call the customer before wasting time on a bad blank.

3. **Roughing with process control for the alloy.** Feeds, speeds, and coolant strategy set for the specific alloy family — sharp coated carbide (or PCD for high-volume Ti Grade 5), positive rake, high-pressure through-tool coolant on 2507 and Inconel to control cutting temperature. Interrupted cuts get extra attention because they cycle heat into the workpiece.

4. **Finish pass to spec.** Finish pass with a fresh insert, controlled DOC, and measured surface-finish targets. Critical features (bore concentricity, seat grooves, sealing surfaces) are held tighter than most job-shop lathes can offer.

5. **CMM verification + documentation.** Every critical dimension CMM-verified. First-article report generated. Certificate of conformance issued with delivery. If your quality group needs a documentation package beyond that, tell us with the RFQ and we'll be straight about what we can and can't produce in-house.

Tolerance and finish expectations in Oil & Gas

±0.0005" routine on critical sealing and mating surfaces. Bore concentricity and roundness held tighter than most job-shop lathes can offer. Surface-finish targets measured with a profilometer, not estimated. For seat grooves and packing bores, finish is often the difference between a sealing part and a leak-path.

Documentation and quality workflow

Material traceability by heat and lot on every job — non-negotiable for API-adjacent work. First-article layout and CMM verification standard. Certificate of conformance issued with delivery. If your quality group needs a documentation package beyond that, send us the requirement with the RFQ and we'll tell you straight whether we can meet it in-house or whether you need a shop with a formal QMS registration.

Response time and emergency work

Rig-down and fleet-down work prioritized over routine production. Same-day and next-day realistic when material is on the shelf. Call (936) 291-7827 direct — say "rig-down" and the phone gets to the shop floor. For emergency work off-hours, leave a voicemail; we route emergencies.

How we compare

Compared to buying finished components from an OEM: we're faster for one-offs and low-volume runs, and we hold traceability the OEM often won't share when you're outside their warranty window. Compared to a commodity job shop: we run the alloys weekly and won't ruin your metallurgy on cutting-temperature mistakes.

Working with B&R Productions

New customer or existing, the process starts the same way: call (936) 291-7827 or send a print through the quote form. Prints are accepted as PDF, STEP, IGES, DWG, or DXF; if you only have a sample part, we reverse-engineer routinely. NDAs on request, and prints stay in-shop rather than going out to third-party vendors. Note for defense and aerospace buyers: B&R is not ITAR- or AS9100-registered, so export-controlled prints and AS9100 flow-down programs need a registered shop — we'll tell you that on the first call, not after the PO.

Quotes typically come back within 24 hours for standard work, same-day for repeat customers on stocked materials, and near-immediately by phone for emergency and rig-down situations. Lead times are quoted honestly — we don't promise Wednesday and deliver next month.

Frequently asked questions

8 questions covering the Hwacheon Mega 100 platform and Oil & Gas work. Also indexed as FAQ schema for AI answer engines.

What's special about the Mega 100's 10" thru-hole?

It accommodates long bar stock, tubing, and hollow parts that most CNC lathes can't clear. Downhole tool bodies with internal passages, coiled-tubing tool bodies, and long valve stems with bore access from both ends — parts most job shops outsource because their machines can't take the stock.

How long a part can the Mega 100 turn?

B&R's stated envelope is 100" (10 ft) between centers. The Mega-100 platform is available up to 6,000 mm (~20 ft) between centers on longer configurations.

What's the chuck swing and max cutting diameter?

Chuck swing over bed 42.5" (1,080 mm); max cutting Ø on the platform 27.5" (700 mm). Two-range spindle at 400/750 RPM with a 25 HP main motor.

What kind of oil & gas customers does B&R Productions work with?

Frac fleet operators, wellhead crews, wireline and coil-tubing service companies, downhole tool manufacturers, completion tool OEMs, and subsea equipment vendors. We work with new customers regularly — first-time orders don't get worse service.

Do you handle rig-down and emergency work?

Yes. Same-day and next-day are realistic when the material is on the shelf (Inconel 718, 625, Super Duplex 2507, 17-4 PH, 4140/4340). Call (936) 291-7827 direct and say "rig-down" — the phone gets to the shop floor, not a form.

Do you machine to API 6A / API 6D tolerances?

Yes — API-adjacent tolerances are routine, and material traceability by heat and lot is standard on every job. Worth being precise about the wording, though: we machine to the tolerances and finishes those specs call for, but B&R is not an API-licensed facility. If your buyer requires parts from an API-licensed shop with a monogram, that's a different vendor. If they require parts machined to print at API-class tolerances with full traceability, that's us.

How do you keep Super Duplex 2507 from losing its chloride resistance during machining?

Controlled cutting temperature — moderate SFM, aggressive feed with sharp coated carbide, high-pressure through-tool coolant. The failure mode nobody wants (sigma-phase intermetallics forming from overheating) shows up months later in service. We run 2507 weekly — this is a process we've internalized, not a spec we're learning on your part.

Can you reverse-engineer discontinued OEM oilfield parts?

Yes — routine work. Send a worn sample and any print or spec you have. We produce the replacement with material traceability the OEM often won't share when you're outside their warranty window.

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Ready to talk about your Oil & Gas work?

Send us a print or a sample photo and specs. Emergencies: call direct and say "rig-down."

Published by B&R Productions — a precision CNC machining shop in New Waverly, Texas, in business since 1994. Serving oil & gas, aerospace, defense, and industrial customers across Texas and the Gulf Coast.

Written by the B&R Productions team. Machine specs verified against manufacturer data. Alloy and process notes reflect shop practice as of 2026-01-15.